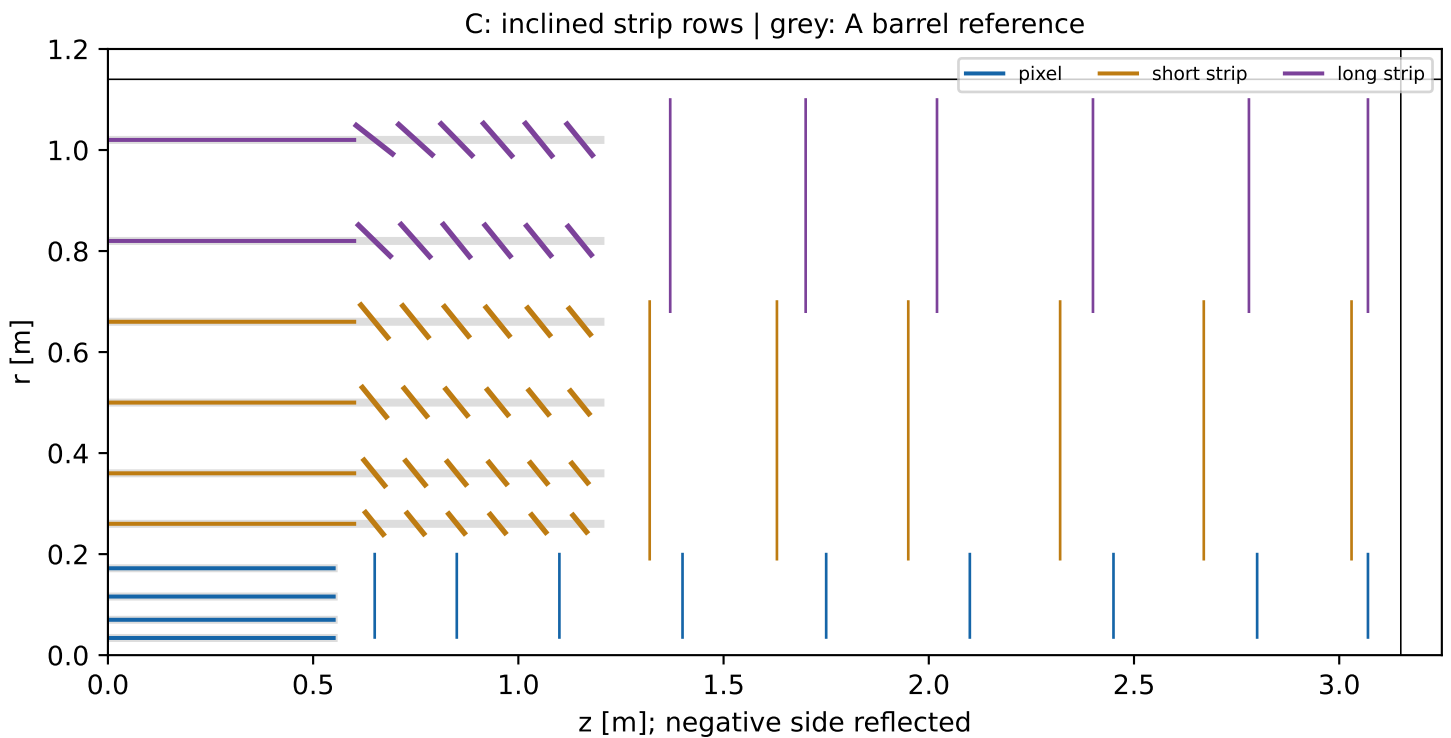


DES-006 C: inclined strip barrel ends

DRAFT / PROTOTYPE | 2026-09-18 | Review response to PR #13

Recommendation: retain C as a targeted material and engineering study alongside baseline A. Its benefit is conditional: less ideal sensor area and lower oblique path length compete with extra row overlaps, routing and assembly complexity. No layout is selected.

Construction: keep A pixels, endcap disks and six strip barrel station radii. Keep each strip barrel flat for $|z| \leq 0.60$ m. Replace each end by six rows centered at $|z| = 0.65, 0.75, 0.85, 0.95, 1.05, 1.15$ m. Tilt each normal toward the origin, capped at 45 degrees. Match origin angular edges, then add 10 mm along each row at both ends.



These conical ring segments approximate the r - z envelope of inclined planar module rows. Phi tiling, active masks, module sizes, finite thickness and actual clearances remain to be designed. The normal material allowance is unchanged; every overlapping row contributes material.

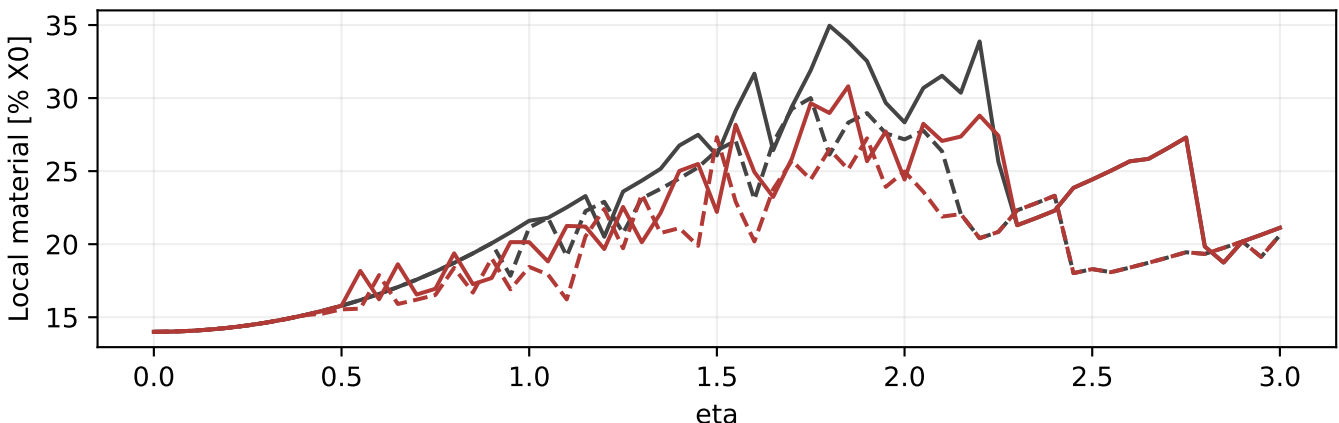
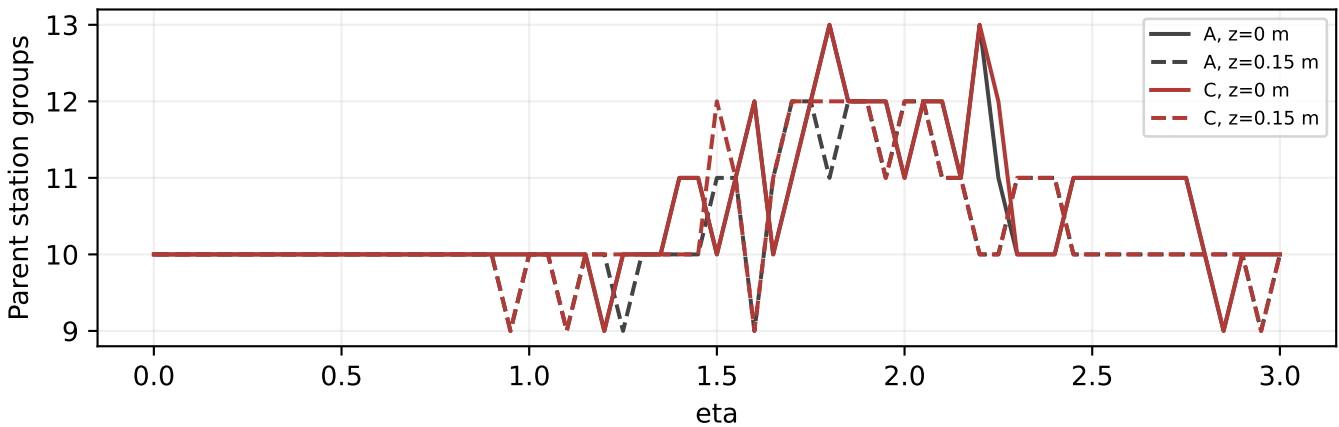
Long strips are double-sided: one stereo module has two scalar sensor faces, one paired station and one 2% X0 local allowance. At $\eta=0$, A and C each give 10 stations, 12 sensor-face crossings and 20 ideal scalar coordinates. Raw IdRes hit counts use a different convention.

Cost-benefit and review decision

Full-system ideal strixel area: A 43.953, C 39.939 m² (-9.1%). Long-strip silicon area counts both faces: A 111.861, C 109.665 m² (-2.0%). Pixel area stays 4.902 m². These are ideal area proxies, not installed module counts, money, power or complete material budgets.

Across 21,627 signed-eta / field / momentum / vertex probes, nominal C loses no parent station groups relative to A. Its local material ratio C/A spans 0.779 to 1.237: overlapping rows can increase material by about 24%. Zero and 5 mm row margins expose gaps; 10 mm is a study allowance, not a qualified clearance.

TrackTech: investigate strixel ends first; the small long-strip area gain may not justify double-sided routing and mechanical complexity. PhysVal: retain both gains and penalties, test tiled coverage before claiming efficiency or lower multiple scattering. C keeps the far-forward pixel system unchanged and does not repair its curvature limitation.



3 T, 100 GeV, local material only. No C IdRes or scattering-fit result.
Detail: DES-006-proposal-C.md; inclined TrackTech/PhysVal inputs; inclined-screen.json.